

internationally visible scientists in 15 STEMM disciplines with at least one journal article indexed in the Scopus database (1992–2021). All had an unambiguously defined gender (binary; male/female, as in the Polish registry of scientists), discipline (in each period), and academic age (i.e., publishing experience). We measured productivity in six-year periods by using publications from individual publishing portfolios and a nonlinear journal prestige-normalized approach. Only the longest working scientists had their productivity calculated in all periods (30 years).

We acted as follows: for every discipline and every period, we ranked all scientists using a full-counting, journal prestige-normalized approach to productivity. Every scientist had a dominant discipline in a period. Dominant disciplines were computed using the modal value from all disciplines (ASJC) assigned to all reference lists accompanying all articles and papers in conference proceedings published by a scientist in a period. We used the ICSR Lab database (run by Elsevier for research purposes) to link all cited references from all publications to individuals. Scientists could be assigned to the same modal discipline in all whole publishing periods or to several modal disciplines in several periods. In this granular way, we avoided a more general approach in which scientists are assigned to a single discipline lifetime. We used the 2-digit (27 subject-level fields) classification instead of the 333 unique fields offered by Scopus because of the small number of observations of top performers per discipline per period. We believe our approach is granular enough to study disciplinary differences.

The gender of scientists was defined using genderize.io, a gender detection tool. We also used academic age calculated as the time passed from the first publication (any type) to any later publication. We defined dominant university affiliations in a given period (binary approach: research-intensive universities of the IDUB type, the rest) using affiliations in all bylines of all publications in a period. The IDUB is a national research excellence program for 2020–2026 for which 10 Polish universities were selected for additional funding. Finally, we studied the productivity of individuals in the context of the productivity of other individuals from the same six-year period (similar opportunities, availability of funding) and the same discipline using productivity classes (10%/90%) rather than publication numbers so that publishing increases and decreases over time did not affect the results.

4. Results

4.1. Bivariate approach

Table 3 presents the share of publications in journals from the 90–99 Scopus percentile ranks, broken down into top performers and others, separately for disciplines and for five consecutive time periods. The group of top performers was defined as the top 10% of authors in a given discipline and period in terms of publishing productivity, which is reflected in its stable share in each cell of the table. The differences between top performers and others were assessed using the Mann-Whitney test, which addresses the hypothesis of convergence of entire distributions; mean values are presented for descriptive purposes only.

In all disciplines and periods, the distribution of the share of publications in the 90–99 percentile ranks for top performers is clearly shifted upward compared to the distribution observed among the remaining authors. This advantage is systematic and (in the vast majority of cases) statistically significant. Lack of significance occurs only sporadically and primarily concerns disciplines with very small numbers in the earliest periods, where the power of the test is limited.

Although the differences between the two classes are presented in the table as mean values, their scale indicates that the top performers' advantage over the rest has clear empirical significance. Authors belonging to the top decile of productivity typically achieve publication shares in top journals that are about two to three times higher than the remaining scientists. In technical and engineering disciplines such as ENG, COMP, ENER, and ENVI, these differences are even stronger, and in the most recent period, the average share of 90–99 percentile publications in the top performer class often exceeds 0.40.

There is strong disciplinary heterogeneity. Technical and engineering disciplines are characterized by higher percentages of articles in the 90–99 percentile ranks in both classes of scientists. In areas such as MATH, MED, and CHEM, absolute values remain lower. Regardless of these structural differences, however, the relative advantage of the top performers over others in distributions remains clear in every discipline.

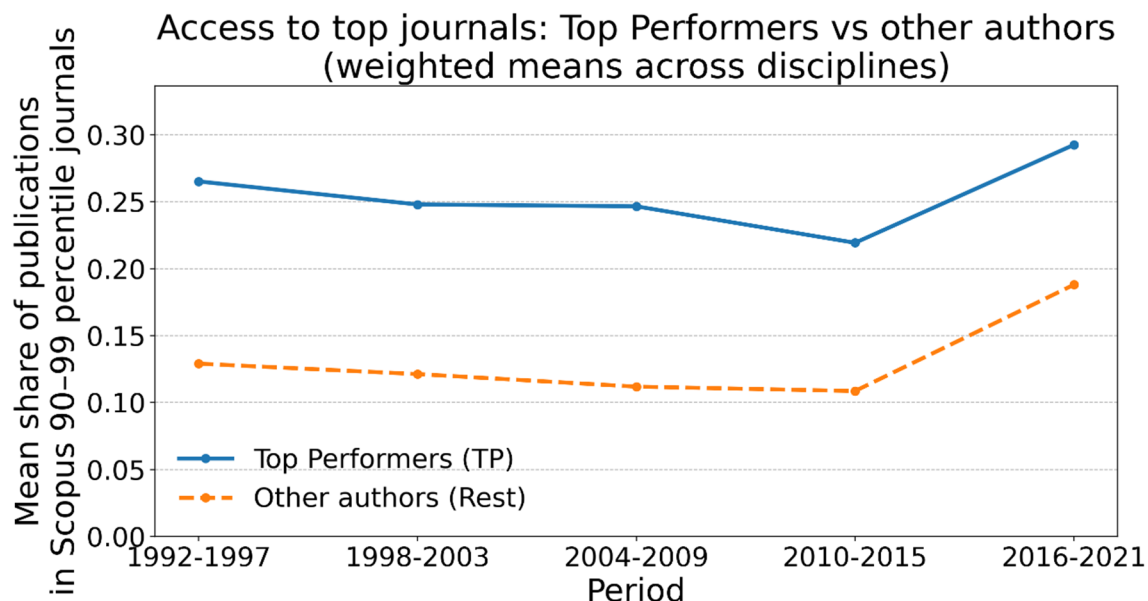
There was an increase in the share of publications in the most prestigious segment (and it was particularly strong in the most recent period of 2016–2021). This increase was observed for both classes of authors, but the difference in the top performers' distributions relative to rest is not weakened. This indicates the strength of mechanisms of productivity stratification over time: despite the powerful expansion of the publication system (and the growing number of articles and scientists), access to top journals remains concentrated in a small group of top performers.

Table 3. Percentage of publications in journals in the 90–99 Scopus journal percentile ranks, top performers vs. rest (Mann-Whitney test)

Period	Discipline	Top performers (upper 10%)	Other scientists (90%)	%_top performers	Mean share of publications in top journals – top performers	Mean share of publications in top journals – other scientists
1992–1997	AGRI	134	1,204	10.0	0.3270***	0.1245
1992–1997	BIO	217	1,945	10.0	0.3140***	0.1437
1992–1997	CHEM	277	2,490	10.0	0.1606***	0.0818
1992–1997	CHEMENG	17	150	10.2	0.2478**	0.1700
1992–1997	COMP	16	138	10.4	0.5115	0.3780
1992–1997	EARTH	70	621	10.1	0.2232***	0.0884
1992–1997	ENER	4	30	11.8	0.8333	0.5000
1992–1997	ENG	82	729	10.1	0.6129***	0.3673
1992–1997	ENVI	33	289	10.2	0.3795***	0.2056
1992–1997	MATER	76	676	10.1	0.3619***	0.2834
1992–1997	MATH	86	770	10.0	0.2618***	0.0871
1992–1997	MED	560	5,046	10.0	0.2163***	0.0971
1992–1997	NEURO	21	181	10.4	0.1477***	0.0765
1992–1997	PHARM	31	274	10.2	0.1620***	0.0528
1992–1997	PHYS	307	2,760	10.0	0.2673***	0.1263
1998–2003	AGRI	251	2,257	10.0	0.2532***	0.1008
1998–2003	BIO	343	3,085	10.0	0.2659***	0.1554
1998–2003	CHEM	432	3,884	10.0	0.1884***	0.1016
1998–2003	CHEMENG	32	280	10.3	0.2413***	0.0926
1998–2003	COMP	29	253	10.3	0.4846***	0.2817
1998–2003	EARTH	130	1,162	10.1	0.2099***	0.0704
1998–2003	ENER	7	73	8.8	0.5675	0.3950
1998–2003	ENG	135	1,207	10.1	0.5041***	0.2790
1998–2003	ENVI	68	610	10.0	0.3674***	0.1465

1998–2003	MATER	116	1,035	10.1	0.3124***	0.2086
1998–2003	MATH	121	1,086	10.0	0.1841***	0.0640
1998–2003	MED	1,039	9,342	10.0	0.2186***	0.0972
1998–2003	NEURO	34	301	10.1	0.1148***	0.0875
1998–2003	PHARM	34	297	10.3	0.1692***	0.0748
1998–2003	PHYS	400	3,592	10.0	0.2715***	0.1321
2004–2009	AGRI	433	3,895	10.0	0.2745***	0.1020
2004–2009	BIO	495	4,452	10.0	0.2671***	0.1670
2004–2009	CHEM	572	5,142	10.0	0.1988***	0.1092
2004–2009	CHEMENG	47	416	10.2	0.2464***	0.0941
2004–2009	COMP	66	591	10.0	0.3292***	0.1571
2004–2009	EARTH	171	1,536	10.0	0.2035***	0.0766
2004–2009	ENER	22	193	10.2	0.5844***	0.1535
2004–2009	ENG	306	2,745	10.0	0.4400***	0.1384
2004–2009	ENVI	131	1,174	10.0	0.3810***	0.1153
2004–2009	MATER	211	1,894	10.0	0.3365***	0.1659
2004–2009	MATH	164	1,469	10.0	0.1834***	0.0647
2004–2009	MED	1,778	15,993	10.0	0.1949***	0.0864
2004–2009	NEURO	42	377	10.0	0.1301***	0.1014
2004–2009	PHARM	37	324	10.2	0.2359***	0.1047
2004–2009	PHYS	473	4,254	10.0	0.2697***	0.1445
2010–2015	AGRI	716	6,442	10.0	0.2448***	0.1006
2010–2015	BIO	702	6,316	10.0	0.2520***	0.1720
2010–2015	CHEM	746	6,707	10.0	0.1751***	0.1124
2010–2015	CHEMENG	48	425	10.1	0.3044***	0.1051
2010–2015	COMP	142	1,271	10.0	0.4103***	0.1349
2010–2015	EARTH	251	2,259	10.0	0.2510***	0.0977
2010–2015	ENER	54	483	10.1	0.4912***	0.1411
2010–2015	ENG	543	4,881	10.0	0.3403***	0.1107
2010–2015	ENVI	247	2,218	10.0	0.3142***	0.1191
2010–2015	MATER	363	3,261	10.0	0.2368***	0.1161
2010–2015	MATH	214	1,921	10.0	0.1717***	0.0707
2010–2015	MED	2,538	22,836	10.0	0.1649***	0.0877
2010–2015	NEURO	62	557	10.0	0.1517***	0.0927
2010–2015	PHARM	47	423	10.0	0.2092***	0.1293
2010–2015	PHYS	563	5,058	10.0	0.2106***	0.1280
2016–2021	AGRI	923	8,300	10.0	0.3361***	0.1944
2016–2021	BIO	884	7,951	10.0	0.3352***	0.2836
2016–2021	CHEM	816	7,341	10.0	0.2511***	0.1648
2016–2021	CHEMENG	56	495	10.2	0.3603***	0.1432
2016–2021	COMP	176	1,581	10.0	0.4608***	0.2799
2016–2021	EARTH	329	2,953	10.0	0.3398***	0.1997
2016–2021	ENER	103	926	10.0	0.6035***	0.4307
2016–2021	ENG	779	7,009	10.0	0.4148***	0.2425
2016–2021	ENVI	384	3,447	10.0	0.4050***	0.2657
2016–2021	MATER	497	4,466	10.0	0.2528***	0.1534
2016–2021	MATH	221	1,987	10.0	0.2703***	0.1181
2016–2021	MED	3,268	29,404	10.0	0.2270***	0.1440
2016–2021	NEURO	88	783	10.1	0.2581***	0.1844
2016–2021	PHARM	53	472	10.1	0.2535***	0.1285
2016–2021	PHYS	603	5,422	10.0	0.2581***	0.1769

Figure 1. Mean percentage of publications in the 90–99 Scopus journal percentile ranks: top performers vs. other scientists (all disciplines combined), by five six-year periods (1992–2021)



The distributions of top performers contrasted with the remaining scientists by discipline and period are shown in Figure 2. Distributions for top performers are consistently shifted toward higher values. They are characterized by higher medians and upper quantiles, while distributions for other authors are strongly concentrated close to zero.

This means that most authors outside the top performer class publish in 90–99 Scopus journals percentile ranks sporadically or not at all, and the differences between the classes become apparent primarily in the upper part of the distribution. Top performers are less diversified than scientists in the other class, suggesting greater stability of presence in prestigious publication channels.

The results show that top performer status is a strong and persistent determinant of access to journals in the 90–99 percentile ranks. The division according to productivity is therefore qualitative and structural in nature. Its mechanisms remain stable over time despite dynamic changes in the intensity of research production.

Figure 2. Share of publications in journals from the 90–99 Scopus journal percentile ranks in terms of top performers (TP) vs. other scientists

